Knowledge Distillation for Edge-Friendly Plant Disease Classification

CENG 463 Final Project Progress Report

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1 Problem Definition

The project studies whether knowledge distillation can produce a compact plant-disease classifier suitable for edge or mobile deployment. The target task is image-based classification on the PlantVillage dataset, where strong teacher models can achieve high accuracy but may be too large or slow for resource-constrained devices. The central research question is whether a MobileNetV3-Small student can approach the performance of larger ResNet teachers while reducing parameter count, model size, and inference latency.

The current report should be interpreted as a technical checkpoint. The implemented experiment uses the three-class PlantVillage Potato subset for fast validation of the data pipeline, model factory, evaluation code, and reporting workflow. These results are not claimed as final full-dataset performance.

2 Dataset and Preprocessing Status

The current experiment uses the PlantVillage Potato subset with three disease or health categories: Early Blight, Late Blight, and Healthy. The subset contains 1,721 RGB leaf images. Because the downloaded dataset does not provide official train, validation, and test splits, the pipeline creates a deterministic random split with seed 42.

Table 1: Dataset split sizes for the Potato subset.

Split	**Samples**	**Fraction**
Train	1,205	70%
Validation	258	15%
Test	258	15%

All images are resized to 224×224 pixels and normalized with ImageNet channel statistics.

Training uses random horizontal flipping and small random rotations, while validation and test transforms use only resizing and normalization. Class labels are remapped after filtering so that the loss function receives contiguous labels from 0 to $C-1$.

3 Literature Review Progress

The literature review currently covers three project-relevant areas. First, Mohanty et al. [1] showed that convolutional neural networks can perform highly accurate image-based plant disease detection on PlantVillage, while also motivating later work on generalization beyond controlled image conditions. Second, Hinton et al. [2] introduced knowledge distillation, where a compact student model learns from both hard labels and softened teacher predictions. The implemented distillation loss is

$$\mathcal{L} = \alpha T^2 \mathrm{KL}\left(\sigma\left(\frac{z_s}{T}\right) \,\Big\|\, \sigma\left(\frac{z_t}{T}\right)\right) + (1-\alpha)\mathrm{CE}(z_s, y), \tag{1}$$

where z_s and z_t are student and teacher logits, T is temperature, α is the distillation weight, and y is the ground-truth label. Third, MobileNet and MobileNetV3 [3, 4] provide efficient architecture choices for edge deployment because they reduce computation through depthwise separable convolutions and mobile-oriented building blocks.

4 Baseline Models

The codebase supports pretrained ImageNet backbones through the `torchvision` weights API. The final classifier layer is replaced to match the number of PlantVillage classes. The current status of the baseline and distillation models is summarized in Table 2.

Table 2: Baseline and distillation experiment status.

Model	**Role**	**Status**
ResNet-18	Initial supervised baseline	Completed on Potato subset
MobileNetV3-Small	Compact supervised baseline	Completed on Potato subset
ResNet-50	Strong teacher candidate	Completed on Potato subset
MobileNetV3 + KD, ResNet-18 teacher	Distilled student	Completed on Potato subset
MobileNetV3 + KD, ResNet-50 teacher	Distilled student	Completed on Potato subset

The implemented ResNet-18 checkpoint uses cross-entropy loss, Adam optimizer, learning rate 10^{-3}, weight decay 10^{-4}, batch size 32, and 3 epochs. The active distillation entry point loads a saved teacher checkpoint, freezes the teacher, and trains MobileNetV3-Small with a hybrid hard-label and soft-target objective. The configuration-driven training code automatically selects CUDA, Apple MPS, or CPU depending on the available hardware.

5 Initial Experimental Results

The current ResNet-18 experiment was read from the saved `resnet18` Potato-subset output directory. It uses a pretrained ResNet-18 with 11,178,051 trainable parameters and evaluates on the 258-image Potato test split. The saved metrics report 98.06% test accuracy, 95.90% macro-F1, 98.00% weighted-F1, and 1.78 ms single-image inference latency on the recorded device. These results are strong for a three-class subset, but they should be treated as an initial pipeline checkpoint rather than evidence of final full-dataset behavior.

Across three epochs, validation accuracy reached 99.61% and validation macro-F1 reached 99.73%, indicating that the training pipeline converged quickly on the Potato subset.

Table 3: Per-class classification report on the test set.

Class	Precision	Recall	F1-score	Support
Potato Early Blight	0.993	1.000	0.996	133
Potato Late Blight	0.962	0.990	0.976	102
Potato Healthy	1.000	0.826	0.905	23
Macro avg	0.985	0.939	0.959	258
Weighted avg	0.981	0.981	0.980	258

Table 4: Summary metrics for the ResNet-18 Potato-subset experiment.

Metric	Value
Test accuracy	98.06%
Test macro-F1	95.90%
Test weighted-F1	98.00%
Trainable parameters	11,178,051
Inference latency	1.78 ms, batch=1
Epochs	3

6 Current Comparison and Technical Direction

The Potato subset now has a complete small-scale comparison across the baseline, teacher, and distilled-student settings. ResNet-50 is the strongest model on this subset, while MobileNetV3-Small remains much smaller. The ResNet-50-to-MobileNetV3 distillation run validates the active KD training pipeline and output generation, but under the current three-epoch default it does not yet outperform the supervised MobileNetV3-Small baseline. This makes temperature, loss weighting, and longer student training the next tuning targets before scaling to the full dataset.

Table 5: Potato-subset comparison results.

Experiment	Scope	Acc.	Macro-F1	Weighted-F1	Params	Latency (ms)
ResNet-18 supervised	Potato subset	98.06%	95.90%	98.00%	11,178,051	1.78
MobileNetV3-Small supervised	Potato subset	94.57%	92.63%	94.57%	1,520,931	3.43
ResNet-50 supervised teacher	Potato subset	99.22%	98.22%	99.22%	23,514,179	4.65
MobileNetV3-Small KD, ResNet-50 teacher	Potato subset	93.41%	88.29%	93.19%	1,520,931	3.54

Current latency measurements on Apple MPS are useful for local tracking but should be repeated on CUDA or CPU under a fixed protocol for a fairer deployment comparison.

7 Ablation and Error Analysis Plan

The ablation plan is designed to isolate the contribution of architecture, distillation, and compression choices:

- Compare the supervised baselines, teacher model, and distilled student under the same Potato-subset split before repeating the comparison on the full 38-class dataset.
- Use the completed ResNet-18 and ResNet-50 teacher runs to compare how teacher strength affects MobileNetV3-Small distillation.
- Vary distillation temperature T and soft-target weight α now that the default KD pipeline is stable.
- Compare supervised MobileNetV3-Small against distilled MobileNetV3- Small to quantify the effect of teacher supervision.
- Measure model size and latency alongside classification metrics; any optional quantization experiment will be treated as a secondary extension after the supervised and KD comparisons are complete.

The error analysis will inspect misclassified test images, per-class precision and recall, confusion matrices, and confidence scores. Special attention will be given to the Potato Healthy class, because its small support makes recall more sensitive to a few errors.

8 Visualization and Interpretation

The current report includes the confusion matrix produced by the ResNet-18 checkpoint. This visualization helps identify whether errors are concentrated between visually similar disease categories or in the underrepresented healthy class.

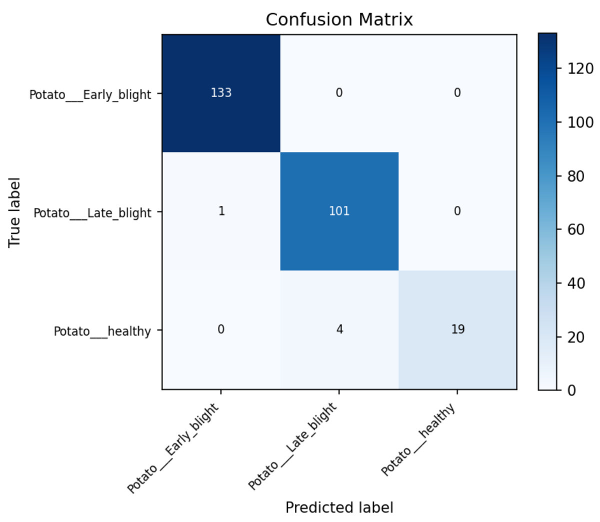

Figure 1: Confusion matrix for the ResNet-18 Potato-subset checkpoint.

Additional visualizations will include training and validation curves, per-class metric bar charts, latency-versus-accuracy plots, and side-by-side examples of correct and incorrect predictions. These will be used to interpret whether a smaller student model fails uniformly or mainly on specific disease classes.

9 GitHub and Reproducibility Status

The project repository is available as a public GitHub repository. The implementation is organized around YAML configuration files, reproducible random seeds, and separate modules for data loading, model construction, training, evaluation, and distillation.

Repository URL:

```
https://github.com/bakiceltik/knowledge-distillation-edge
```

The active Potato-subset experiments can be reproduced with:

```
python -m src.data --config configs/baseline_resnet18.yaml
python -m src.train_baseline --config configs/baseline_resnet18.yaml
python -m src.train_baseline --config configs/student_mobilenetv3.yaml
python -m src.train_baseline --config configs/teacher_resnet50_potato.yaml
python -m src.train_distillation --config configs/distillation_resnet50_potato.
    yaml
```

The saved checkpoint artifacts used in this report are the metrics JSON, results CSV, training-history CSV, model-summary JSON, and confusion-matrix PNG files from the Potato-subset output directories.

10 Current Challenges and Next Steps

The main current challenge is that the Potato subset is small and visually controlled, so it may overestimate performance compared with the full 38-class PlantVillage task and real-world field images. The healthy Potato class is also much smaller than the disease classes, which makes class-level recall less stable. Latency measurement is another challenge: the current MPS timings are useful for local tracking but should not be treated as final hardware-neutral latency results.

The immediate next step is no longer basic KD validation; that path now runs end-to-end with a saved ResNet-50 teacher and produces the expected artifacts. The next research step is to tune distillation temperature, the hard-label and soft-target loss balance, and the number of student epochs, because the current ResNet-50 distillation run is below the supervised MobileNetV3-Small baseline. After the subset experiments are tuned, the pipeline will be scaled to the full 38-class PlantVillage dataset. Quantization remains a secondary extension after the supervised and KD comparisons are stronger.

References

[1] S. P. Mohanty, D. P. Hughes, and M. Salathe, "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.

[2] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," *NIPS Deep Learning Workshop*, 2015.

[3] A. G. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," *arXiv:1704.04861*, 2017.

[4] A. Howard et al., "Searching for MobileNetV3," *IEEE/CVF ICCV*, pp. 1314–1324, 2019.